BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 8485/8488

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Chemistry - Paper: II Objective (iii)

کیمسٹری

Time Allowed: 20 Minutes	**Maximum Marks:** 17	**Paper Code:** 8485

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Q1. Acetic acid is prepared by:

(A) Distillation
(B) Fermentation
(C) Ozonolysis
(D) Esterification

Q2. Ketones are prepared by oxidation of:

(A) Primary alcohol
(B) Secondary alcohol
(C) Tertiary alcohol
(D) Ether

Q3. Which compound shows the hydrogen bonding?

(A) C_2H_6
(B) C_2H_5Cl
(C) CH_3-O-CH_3
(D) C_2H_5OH

Q4. The p^H range of acid rain is:

(A) 7-6.5
(B) 6.5-6
(C) 6-5.6
(D) Less than 5

Q5. The co-agulant used in raw water to precipitate suspended impurities is:

(A) Caustic soda
(B) Lime water
(C) Alum
(D) Soda ash

Q6. During the S.1 reaction, the fast step involves:

(A) Breakage of covalent bond
(B) Formation of carbocation
(C) Transition state
(D) Attack of nucleophile

Q7. Most of the elements of group I A are:

(A) Crystalloids
(B) Metals
(C) Metalloids
(D) Non metals

Q8. In group V-A elements, the most electronegative is:

(A) [illegible in source scan]
(B) P
(C) Sb
(D) Bi

Q9. During nitration of benzene, the active nitrating agent is:

(A) NO_3
(B) NO_2^-
(C) NO_2^+
(D) HNO_3

Q10. Vinyl acetylene combines with HCl to form:

(A) Poly acetylene
(B) Benzene
(C) Chloroprene
(D) Divinyl acetylene

Q11. The state of hybridization of carbon atom in methane is:

(A) sp^3
(B) sp^2
(C) sp
(D) dsp^2

Q12. Total number of transition elements are:

(A) 10
(B) 14
(C) 40
(D) 58

Q13. Which is the strongest acid?

(A) HClO
(B) $HClO_2$
(C) $HClO_3$
(D) $HClO_4$

Q14. Which element belongs to group IV A of periodic table?

(A) Barium
(B) Iodine
(C) Lead
(D) Oxygen

Q15. Which one of the given is not an alkali metal?

(A) Francium
(B) Caesium
(C) Rubidium
(D) Radium

Q16. For which crop, ammonium nitrate fertilizer is not used:

(A) Cotton
(B) Wheat
(C) Sugarcane
(D) Paddy rice

Q17. Vegetable oil is:

(A) Un-saturated fatty acid[illegible in source scan]
(B) Glycerides of unsaturated acid
(C) Glycerides of saturated fatty acid
(D) Essential oils obtained from plants

Chemistry - Paper: II Subjective

کیمسٹری

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Q2.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Why the size of an anion is greater than its parent atom? Give example also.

(ii) The hydration energies of the ions are in the given order, prove it: $Al^{3+} > Mg^{2+} > Na^{+}$.

(iii) Why is potassium superoxide used in breathing equipments?

(iv) How is gypsum converted into Plaster of Paris?

(v) Write down the systematic names of the given: a) $[Fe(CO)_5]$, b) $K_2[PtCl_6]$.

(vi) Write down the structure of $Cr_2O_7^{-2}$ and MnO_4^{-1}.

(vii) Define β-elimination reactions with a suitable example.

(viii) How anti-knocking agents are prepared from methyl and ethyl chloride?

(ix) What is denaturation of proteins?

(x) Define isoenzymes.

(xi) Differentiate between thermosetting and thermoplastic polymer. Give example.

(xii) What do you mean by "setting of cement?"

Q3.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) How PCl_3 reacts with H_2O and C_2H_5OH?

(ii) Why does aqua regia dissolve gold?

(iii) Why HF is weaker acid than HCl?

(iv) What are Freons and Teflon?

(v) What is "Octane number" and "knocking"?

(vi) What is carbonization of coal?

(vii) Which rule is followed by alkenes to addition of H_2SO_4 in 1-Butene?

(viii) Prepare Cis and trans-butene from 2-butyne.

(ix) Write down the IUPAC names of the given compounds: a) $(C_6H_5)_2CH_2$ b) $(CH_3CH_2CH_2)_3CH$

(x) What are leachates?

(xi) What are the advantages of recycling of paper? (in two points)

(xii) What is acid deposition?

Q4.. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Write down chemistry of borax bead test.

(ii) Why is nitric acid frequently transported in aluminium containers?

(iii) CO_2 is a gas while SiO_2 is a solid. Give reason.

(iv) What is Wurtz-Fitting reaction?

(v) What is denaturing of alcohols?

(vi) What is Williamson's synthesis of ether?

(vii) What is formalin? How is it prepared in laboratory?

(viii) How is acetic acid prepared by hydrolysis of ester?

(ix) Elaborate acidic and basic character of amino acids.

> Note: SECTION-C Note: Attempt any THREE questions. Each Question carries EIGHT (8) marks. (8x3=24)

Q5..

(a) [4] What are the improvements made in the Mendeleeve's periodic table?

(b) [4] Describe with diagram, the manufacture of sodium by Down's Cell method.

Q6..

(a) [4] Why Fluorine shows peculiar behaviour? (Any four reasons)

(b) [4] Describe briefly the given steps during the preparation of paper: i-Cleaning ii-Screening

Q7..

(a) [4] What is orbital hybridization? Explain sp^3 hybridization with an example.

(b) [4] How does ethyl magnesium iodide react with: i. CO_2 ii. CH_3CHO iii. H_2O iv. C_2H_5OH

Q8..

(a) [4] How will you prepare alkynes by Kolb's electrolysis?

(b) [1+3] How does acetaldehyde react with $NaBH_4$? Give mechanism of reaction.

Q9..

(a) [4] Write down a note on Atomic orbital treatment of benzene.

(b) [2+2] Mention any two reactions of alcohol in which C-O bond is broken and any two reactions of alcohol in which O-H bond is cleared.

GROUP – II

Chemistry - Paper: II Objective (iv)

کیمسٹری

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8488

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Q1. During nitration of benzene, the reactive nitrating agent is:

(A) NO_3 (B) NO_2^+

(C) NO_2 (D) HNO_3

Q2. The presence of a double bond in a compound is the sign of:

(A) Saturation (B) Unsaturation

(C) Substitution (D) None of these

Q3. In t-butyl alcohol, the tertiary carbon is bonded to ------- hydrogen atom/s.

(A) 2 (B) 3

(C) 1 (D) No

Q4. The colour of transition metal complexes, is due to:

(A) d-d transition of electrons
(B) Ionization
(C) Paramagnetic nature
(D) Loss of S-electrons

Q5. Boric Acid can not be used:

(A) As antiseptic in medicine
(B) For washing eyes
(C) In soda bottles
(D) For enamels and glazes

Q6. The brown gas is formed, when metal reduces HNO_3 to:

(A) N_2O_5
(B) N_2O_3
(C) NO_2
(D) NO

Q7. The anhydride of $HClO_4$ is:

(A) ClO_3
(B) ClO_2
(C) [illegible in source scan]
(D) Cl_2O_7

Q8. Which of the given sulphates is not soluble in water?

(A) Sodium sulphate
(B) Potassium sulphate
(C) Zinc sulphate
(D) Barium sulphate

Q9. Na forms ------- oxide:

(A) Basic
(B) Amphoteric
(C) Acidic
(D) Super

Q10. Ecosystem is a smaller unit of--------

(A) Biosphere
(B) Atmosphere
(C) Lithosphere
(D) Hydrosphere

Q11. The value of ------- is an indirect measure of chemically oxidizable matter in water.

(A) COD
(B) BOD
(C) DO
(D) None of these

Q12. The nitrogen present in some fertilizers helps plants:

(A) To fight against diseases
(B) To produce fat
(C) To produce protein
(D) To undergo photosynthesis

Q13. Which of the given polymers is a synthetic polymer?

(A) Animal fat
(B) Starch
(C) Cellulose
(D) Polyester

Q14. Acetic acid is manufactured by:

(A) Distillation
(B) Fermentation
(C) Ozonolysis
(D) Esterification

Q15. Which of the given compounds will not give iodoform test on treatment with I_2/NaOH?

(A) Acetaldehyde
(B) Acetone
(C) Butanone
(D) 3-pentanone

Q16. Methyl alcohol is not used:

(A) As a solvent
(B) As an anti-freezing agent
(C) As a substitute for petrol
(D) For denaturing of ethyl alcohol

Q17. Order of a typical S.N2 reaction in case of primary alkyl halide is:

(A) 1
(B) 2
(C) [illegible in source scan]
(D) Zero

Chemistry - Paper: II Subjective

کیمسٹری

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Note: Note: Section B is compulsory. Attempt any 3 questions from Section C.

Q2.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Why diamond is a non conductor and graphite is a good conductor?

(ii) The hydration energy of ions is in the given order, $Al^{3+}>Mg^{2+}>Na^{+1}$, give reason.

(iii) What is milk of magnesia and what is its use?

(iv) What is Gypsum and how is it converted into Plaster of Paris?

(v) Why melting and boiling points are maximum in the middle of the series of d-block element?

(vi) What is the reason for the development of colours in the compound of transition elements?

(vii) How alkyl iodide is prepared from alcohol?

(viii) What are the factors which are responsible for the reactivity of alkyl halide?

(ix) What is degree of polymerization? Give one example.

(x) Define Homopolymer and Copolymer; give example for each polymer also.

(xi) What are epoxy resins? Give their important uses.

(xii) Write down essential qualities of a good fertilizer.

Q3.. (Marks 16) Write short answers to any EIGHT parts. (8x2=16)

(i) Draw structure of two oxyacids of nitrogen.

(ii) Why does the phosphorous show more than one valency although it is not a transition element?

(iii) Who do the noble gases have low melting and boiling points?

(iv) Give two methods to prepare ClO_2.

(v) What is catalytic cracking? Give its importance.

(vi) What are Cis. and trans isomers? Give example.

(vii) Why is the boiling point of n-butane higher than that of isobutane?

(viii) Why alkanes are less reactive than alkenes?

(ix) Give two methods to prepare alkanes from alkyl halides.

(x) What are secondary pollutants? Give example.

(xi) How are oxides of sulphur produced in environment? Give their harmful effects in human life.

(xii) Define oxidizing and reducing agent.

Q4.. (Marks 12) Write short answers to any SIX parts. (6x2=12)

(i) Write down any four uses of aluminium.

(ii) What do you know about chemical garden?

(iii) How does borax serve as water softening agent?

(iv) Convert benzene into acetophenone.

(v) Write down structural formulas for: (a) Glycol (b) Carbolic Acid

(vi) How will you distinguish between methanol and ethanol?

(vii) Give one industrial method for the preparation of formaldehyde.

(viii) Write down a short note on acidic and basic characters of amino acid.

(ix) Convert methyl nitrile into Acetic Acid.

> Note: SECTION-C Attempt any THREE questions. Each Question carries EIGHT (8) marks. (8x3=24)

Q5..

(a) [4] Describe the defects in Mendeleev's periodic table. Give two improvements made in it by Mosley.

(b) [4] Discuss the trends in chemical properties of alkaline earth metals (any four).

Q6..

(a) [4] Write down any four differences of F_2 with its group members.

(b) [4] Define the term "setting of cement". Write down the reactions involved in first 24 hours and 1-7 days.

Q7..

(a) [4] What is orbital hybridization? Explain sp^2-hybridization in ethene.

(b) [4] Differentiate between S_N1 and S_N2 reaction mechanisms.

Q8..

(a) [4] Write down any two methods for the preparation of alkanes.

(b) [4] Give mechanism for acid catalysed nucleophilic addition reaction of ethanal and propanone with hydroxylamine.

Q9..

(a) [4] What is Friedel-crafts acylation? Write down its mechanism for the preparation of acetophenone.

(b) [4] Discuss the commercial preparation of methyl alcohol from water gas in detail.